

Virtual Physics Laboratory Suite

SURE Project Report

Creating Interactive Web Based Physics Experiments

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1 Introduction

The main goal of this internship was to build a set of interactive online physics labs that students can use to play with experiments in real time. Most standard digital learning tools use simple videos or basic animations that do not follow true physics laws. This project fixes that problem by using step by step math equations to make sure every particle moves naturally.

The project features four distinct labs covering different areas of physics, including electrical fields, magnets, floating oil drops, and bouncing shapes. Every single movement and collision is calculated live using exact physics formulas. This report explains how these four labs were built, what tools were used, and how we fixed everyday programming and layout challenges during development.

2 Technologies and Dependencies

To make sure the labs open instantly on any device without downloading heavy software, everything runs completely inside the user web browser. The setup does not require any backend servers or complicated code frameworks.

For the core tech stack, standard HTML5 and CSS3 were used to design the layout and control panels. This ensures that the sliders and buttons stay locked in place and look clean on different screens. Plain JavaScript handles the underlying math, page logic, and user settings.

For additional help, we used the p5.js library to manage the visual canvas, draw smooth graphics at sixty frames per second, and process positional math. The Bootstrap 5 library was used to quickly build clean menus, sliders, and buttons, while MathJax was added to display clean mathematical symbols directly on the web page.

3 Simulation 1: Rutherford Alpha Scattering

3.1 Overview

The first lab models the famous gold foil experiment which shows how matching electric charges push away from each other. The code calculates the continuous pushing force between moving positive alpha particles and a heavy atom core.

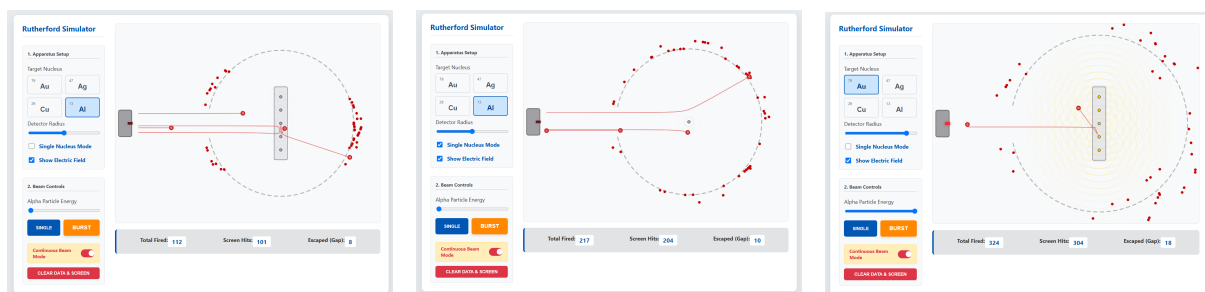
3.2 Features

Students can switch between different target metals like Gold, Silver, Copper, and Aluminum to see how a larger or smaller atomic charge changes the paths of the particles. The lab lets users fire single particles, launch quick bursts, or turn on a steady stream, while a circular screen counts where the particles land.

3.3 Challenges and Solutions

When firing a continuous stream of particles, the web browser would quickly slow down because thousands of old particles stayed hidden in computer memory. We fixed this by rewriting the cleanup code to check the list of particles backward, cleanly deleting any object that left the screen area.

Another issue was that drawing the large glowing fields around heavy elements like Gold caused the screen to stutter. We solved this by rendering the static background glows onto a hidden canvas sheet once, then stamping that image on the main screen to keep the animation smooth.



(a) Metal Panel Mode

(b) Single Nucleus Mode

(c) Electric field lines ON

Figure 1: Rutherford Alpha Scattering Simulation Environment

4 Simulation 2: J.J. Thomson Cathode Ray Tube

4.1 Overview

This lab recreates the experiment that discovered the electron. It simulates a stream of electrons moving through a glass tube while being bent by an electric plate and a magnetic coil.

4.2 Features

The simulator provides two viewing angles, allowing users to switch between a side view of the glass tube and a front view of the circular target grid. It also includes a built in calculator where students can type in their slider values to figure out the electron properties and see their error percentage compared to real world data.

4.3 Challenges and Solutions

The initial canvas window was too wide for standard laptop screens, which caused the right border of the lab to get cut off. We fixed this by shrinking the canvas width to nine hundred and eighty pixels and shifting the central starting point fifty pixels to the left so everything stays perfectly centered. We also removed the harsh container borders so the workspace blends cleanly with the rest of the page.

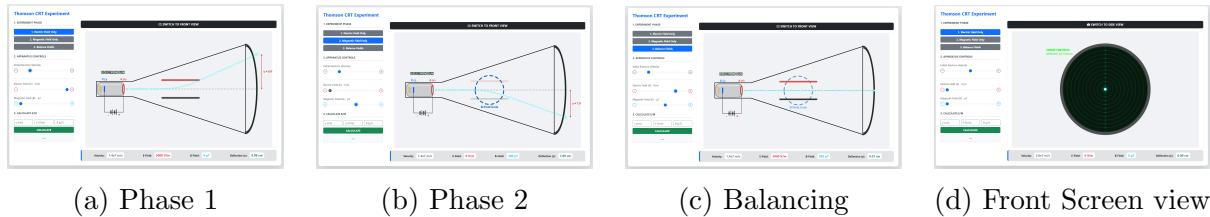


Figure 2: J.J. Thomson Cathode Ray Tube Simulation Framework

5 Simulation 3: Millikan Oil Drop Experiment

5.1 Overview

This lab models the experiment used to measure the charge of a single electron. It controls the movement of tiny oil drops by balancing gravity, air buoyancy, electrical fields, and air resistance.

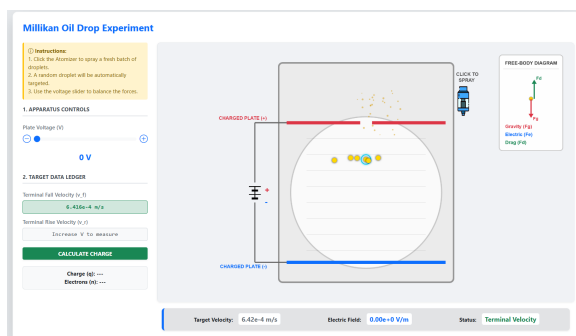
5.2 Features

The lab features a spray button to launch a batch of random oil droplets, an automated data logger that locks in the falling and rising speeds without human timing errors, and a live force diagram that changes size based on the forces acting on the selected drop.

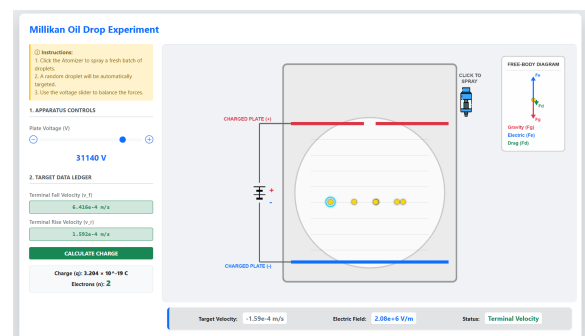
5.3 Challenges and Solutions

Loading the atomizer graphic inside the main drawing loop caused depth errors and blocked the falling drops. We fixed this by moving the sprayer out of the canvas drawing layer entirely and placing it as a standard HTML image overlay using absolute positioning.

Additionally, spraying a new batch of drops while the electric voltage was already turned up caused the new drops to fly off the screen instantly. We resolved this by forcing the voltage slider back to zero every time the spray button is clicked, resetting the environment for the new droplets.



(a) Atomizer Chamber & Droplets



(b) Terminal Balance & FBD Analysis

Figure 3: Millikan Oil Drop Calibration Lab

6 Simulation 4: Solid Sphere Kinematic Scattering

6.1 Overview

The final module focuses on mechanical collisions. It works as a physics puzzle where users fire small beams at solid shapes like circles, squares, and triangles to see how they bounce.

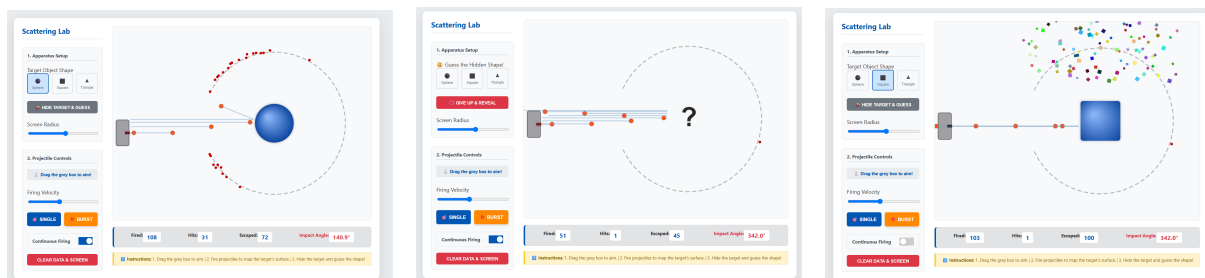
6.2 Features

The lab has a launcher box that users can drag up and down to aim. It includes a secret mode where the target shape is hidden from view, forcing the student to figure out what shape is in the middle by looking at how the lines bounce off it.

6.3 Challenges and Solutions

At high speeds, the fired particles would glitch and jump straight through the triangle walls in a single frame, getting trapped inside. We fixed this by dividing the movement calculation into four smaller micro steps per frame, making it impossible for a particle to skip past a boundary line.

We also noticed the randomizer would sometimes pick the exact same hidden shape twice in a row, making the puzzle feel repetitive. We added a loop that checks the previous selection and forces the game to choose a completely different shape every time you restart. Finally, a drawing glitch caused shapes to fade away unexpectedly, which we fixed by using strict separation commands for each graphic layout.



(a) Sphere Scattering Map

(b) Mystery Shape Collision

(c) Guessing the shape

Figure 4: Solid Object Geometric Kinematic Scattering UI

7 Conclusion

Building the Virtual Physics Laboratory Suite shows how clean web programming can make science education much more interactive. Over this internship, four standalone physics engines were created from scratch.

The project required a clear understanding of state management, tracking coordinates, vector math, and updating web pages smoothly. By making sure everything runs directly in the browser without extra downloads, the labs are fast and easy for anyone to open. Working through problems like particles clipping through walls and screen slowdowns gave me great experience in optimizing web code. The final product successfully shows how standard web tools can make abstract physics principles easy to see and understand.